

Growth Theory (Part 2)

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus du Havre

Lectures 3

Announcements

- Textbook: Chapter 12

Solow model with population growth and technological progress

- Previous lectures (L2): we assumed no population growth and no technological progress
- Today's lecture: we relax these assumptions and we revisit the Solow model

Technological progress

Technological progress can lead to:

- larger quantities of output for given quantities of capital and labor
- better products
- new products
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Production function

$$Y = F(K, AN)$$

where AN is the amount of effective labor.

Properties of the new production function

...are the same as before:

1. Constant returns to scale (CRS)

- If we double **all** inputs $\implies$ the output doubles:

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2. Positive and diminishing marginal products

- **Decreasing returns to capital:** Increases in capital lead to smaller and smaller increases in output.
- **Decreasing returns to *effective* labor:** Increases in effective labor lead to smaller and smaller increases in output.

Notes on notation

In the previous lectures, we simplified the notation:

small letters denoted “per worker” variables:

$$y = \frac{Y}{N}$$

$$k = \frac{K}{N}$$

Today, we will introduce “per effective worker” variables:

$$\hat{y} = \frac{Y}{AN}$$

$$\hat{k} = \frac{K}{AN}$$

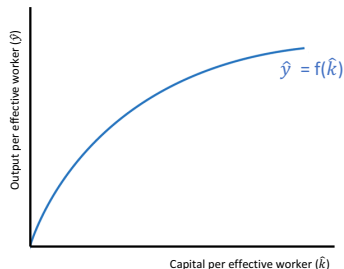
Simplified production function

$$Y = F(K, AN)$$

$$\frac{Y}{AN} = F\left(\frac{K}{AN}, 1\right)$$

$$\hat{y} = f(\hat{k})$$

Output per effective worker is a function
of capital per effective worker



Solow model

The law of motion of capital:

$$\text{Change of capital} = (\text{actual investment}) - (\text{break-even investment})$$

Now, everything will be “per effective worker”

Solow model

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1. Actual investment

- Recall: $I = S = sY$
- Divide everything by effective labor (AN):
- Actual investment per effective worker is defines as:

$$\frac{I}{AN} = sf\left(\frac{K}{AN}\right) = \mathbf{sf}(\hat{\mathbf{k}})$$

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2. Break-even investment

- Level of investment per effective worker needed to maintain a given level of capital per effective worker because

$$\hat{i} = (\delta + g_A + g_N) \hat{k}$$

where

- δ is depreciation rate
- g_A is the rate of technological progress
- g_N is the rate of population growth

Solow model

Change of capital = actual investment – break-even investment

$$\Delta \hat{k} = sf(\hat{k}) - (\delta + g_A + g_N)\hat{k}$$

Solow model

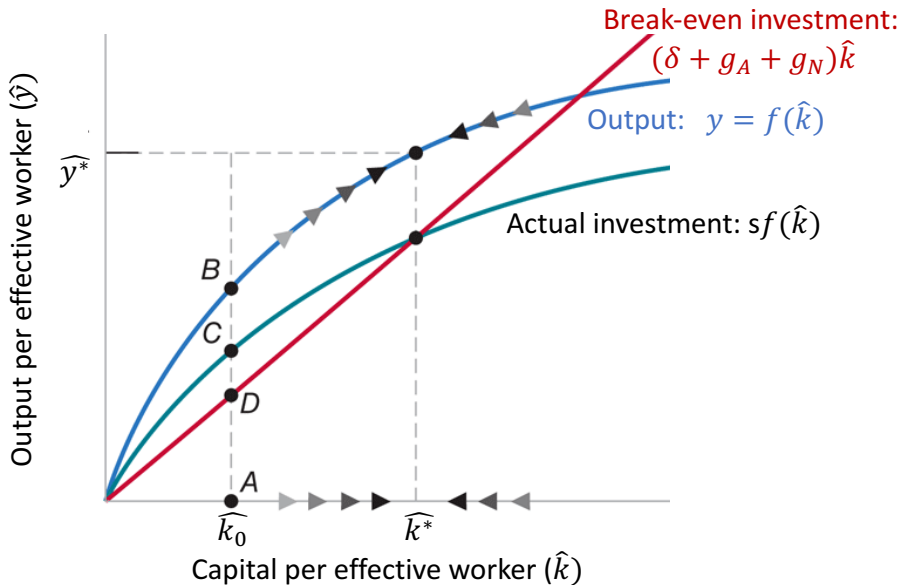
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Dynamics

- if $sf(\hat{k}) = (\delta + g_A + g_N)\hat{k} \implies \Delta \hat{k} = 0$ **steady state**
- if $sf(\hat{k}) > (\delta + g_A + g_N)\hat{k} \implies \Delta \hat{k} > 0$ capital accumulation
- if $sf(\hat{k}) < (\delta + g_A + g_N)\hat{k} \implies \Delta \hat{k} < 0$ capital losses

Solow model



The Characteristics of Balanced Growth

Growth rates		
Capital per effective worker	$\hat{k}$	???
Output per effective worker	$\hat{y}$	???
Labor	N	???
Capital	K	???
Output	Y	???
Capital per worker	k	???
Output per worker	y	???

The Characteristics of Balanced Growth

		Growth rates
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Output	Y	???
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On the balanced growth path (steady state / long run):

Capital per effective worker and output per worker are constant. ($\Delta \hat{k} = 0$, $\Delta \hat{y} = 0$)

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On the balanced growth path (steady state / long run):

Population grows at the population growth rate

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Output	Y	$g_A + g_N$
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On the balanced growth path (steady state / long run):

Capital and output are growing at a rate equal to the sum of population growth and the rate of technological progress.

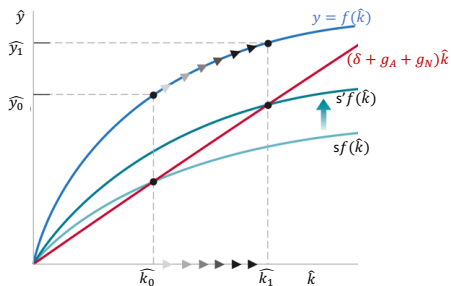
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Capital per worker	k	g_A
Output per worker	y	g_A

On the balanced growth path (steady state / long run):

Capital per worker and output per worker are growing at the rate of technological progress

The Effects of an Increase in the Saving Rate

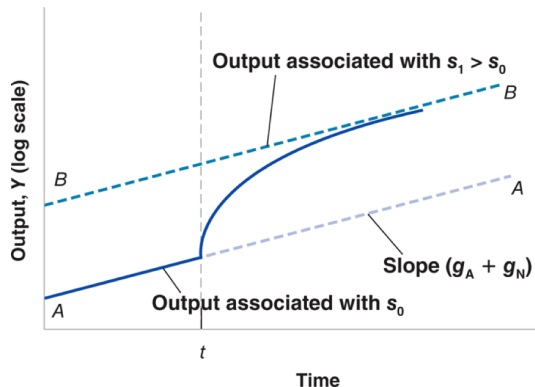


An increase in the saving rate:

- Shifts actual investment curve up
- Leads to an increase in the state-state levels of output per effective worker and capital per effective worker

Convergence

The increase in the saving rate leads to higher growth until the economy reaches its new, higher, balance growth path



Constructing a Measure of Technological Progress

- Can we actually measure technological progress it empirically?
- In 1957, Robert Solow devised a way to construct an estimate of technological progress

Solow residual (derivation)

Let's take Cobb-Douglas production fcn:

$$Y = K^{1-\alpha}(AN)^{\alpha}$$

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Take a log at time t

$$\log(Y_t) = (1 - \alpha)\log(K_t) + \alpha\log(A_t) + \alpha\log(N_t) \quad (1)$$

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Compute the difference: (1) - (2):

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Rearranging

$$\alpha g_A = g_Y - (\alpha g_N + (1 - \alpha)g_K)$$

Solow residual

$$\text{residual} = g_Y - (\alpha g_N + (1 - \alpha)g_K)$$

- From data, we know the growth of output (g_Y) and labor (g_N). Applying some simple assumptions, we can compute the growth of capital (g_K)
- From data, we also know the share of labor (α) and we can derive the share of capital ($1 - \alpha$)
- The Solow residual measures the effects of technological progress
- It is also known as a **rate of growth of total factor productivity** (or the **rate of TFP growth**)
- The Solow residual is also equal to the share of labor times the rate of technological progress: $\text{residual} = \alpha g_A$